Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

4th Year 1st Semester Final Examination-2022

Course Code: EEE 455 Session: 2018-19

Course Title: VLSI Design and Testing

Total Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part - A

(Answer any two of the following questions)

1. (a) What are the effects of O_2 and H_2 adding on etching? 5

   (b) Discuss the silicon oxidation model mentioning its limiting cases. 5

   (c) Explain how does the plasma sustain. Also explain the chemistries involved in plasma etching. 7

   (d) Write the VHDL code to implement the following Boolean expression: $F = AB + \overline{A}BC + BC$. 8

2. (a) Find threshold voltage of MOSFET without body effect. 6

   (b) State Moore's law. Explain how different parameters of MOSFET is scaled. 6

   (c) Explain operation of saturated load NMOS inverter. What are the disadvantages of this configuration? 5

   (d) Find DC characteristics of CMOS inverter using V-I characteristics curve. 8

3. (a) What is VHDL? Write down the advantages of VHDL. 2

   (b) Sketch a 3-input NOR gate with transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model. Assume that every source or drain has fully contacted diffusion when making your estimate of capacitance. 6

   (c) For the circuit in figure 3c find- 10

   i) Logical effort, G
   ii) Branching effort, B
   iii) Electrical effort, E
   iv) Path effort, F
   v) Stage effort, f
   vi) Parasitic delay, P
   vii) Minimum path delay, D
   viii) Size of the gate.

   (d) Find the number of stages for minimum delay. 7

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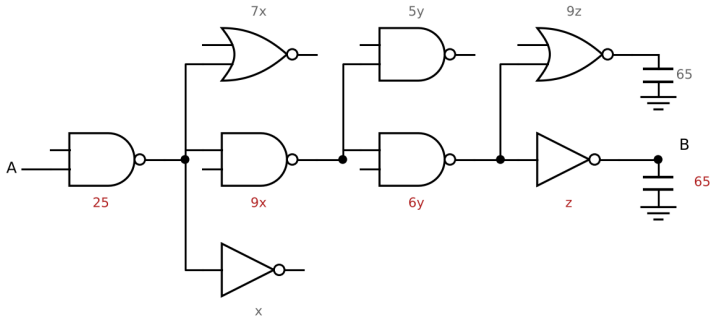


Figure: 3c of the paper (redrawn); the dashed path from A to B on the paper runs through the gates marked 25, 9x, 6y and z

Part - B

(Answer any two of the following questions)

4. (a) Explain why NMOS network is not used as pull-up network. 5

   (b) What are the sources of power dissipation in CMOS logic circuit? 5

   (c) What is activity factor? Figure 4.1 shows a 4-input AND gate built using a tree and a chain of gates. Determine the activity factors at each node in the circuit assuming the input probabilities PA = PB = PC = PD = 0.5. 7

   (d) A digital system-on-chip in a 1 V 65 nm process (with 50 nm drawn channel lengths and $\lambda = 25$ nm) has 1 billion transistors, of which 50 million are in logic gates and the remainder in memory arrays. The average logic transistor width is 12 λ and the average memory transistor width is 4 λ. The memory arrays are divided into banks and only the necessary bank is activated so the memory activity factor is 0.02. The static CMOS logic gates have an average activity factor of 0.1. Assume each transistor contributes 1 fF/Rm [*sic*: should be fF/µm] of gate capacitance and 0.8 fF/µm of diffusion capacitance. Neglect wire capacitance for now (though it could account for a large fraction of total power). **Estimate the switching power when operating at 1 GHz.** If Subthreshold leakage for OFF devices is 100 nA/µm for low-threshold devices and 10 nA/µm for high-threshold devices. Gate leakage is 5 nA/µm. Junction leakage is negligible. Memories use low-leakage devices everywhere. Logic uses low-leakage devices in all but 5% of the paths that are most critical for performance. **Estimate the static power consumption.** 8

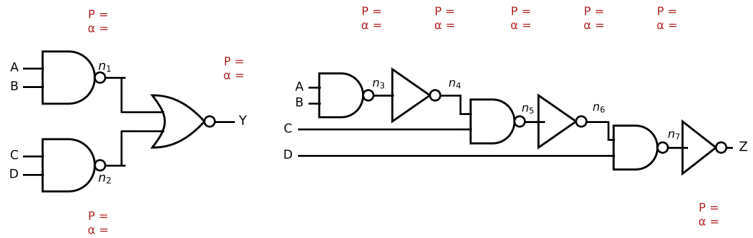


Figure: 4.1 of the paper (redrawn): the 4-input AND as a tree and as a chain

5. (a) Draw the stick diagram of the following Boolean function: $f = \overline{A.B.C}$ with their symbolic diagram and truth table. 8

   (b) Write down the advantages of CMOS. 5

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(c) Explain max-delay constraint and min-delay constraint in sequential circuits. 6

(d) Sketch logic circuit of flip-flop with synchronous reset and explain its operation. 6

6. (a) Explain the principle of dynamic inverter. 5

(b) Design and implement full adder logic circuit. 6

(c) Define PLA and PAL. Sketch a 3-input and 2-output PLA implementing the following logic expressions- $s = a\overline{b}\,\overline{c} + \overline{a}b\overline{c} + \overline{a}\,\overline{b}c + abc$, $c = ab + bc + ac$ 7

(d) Explain how data is written and read in a DRAM sub-array. 7

Term Test 1 (its questions 1 to 3 are 7(a) to 7(c))

[4th Year 1st Semester Term Test-01, Set - A, Full Marks: 20, Time: 40 minutes]

7. (a) Briefly describe steps of photo-lithography. 8

(b) Explain the silicon oxidation model mentioning its limiting cases. 6

(c) Briefly explain vapor phase epitaxy. 6

Term Test 2 (its questions 1 to 5 are 8(a) to 8(e))

[4th Year 1st Semester Term Test-02, Set - B, Answer any four out of five questions, Full Marks: 20, Time: 40 minutes]

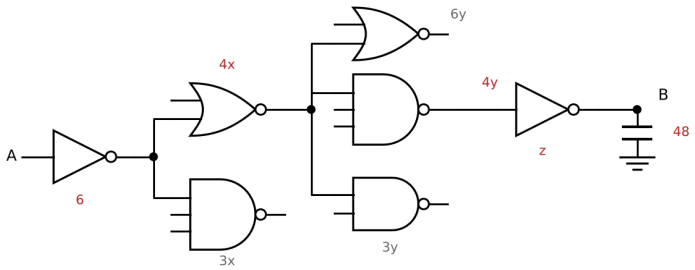
8. (a) Find the threshold voltage of MOSFET considering the body effect. 5

(b) Briefly explain the following terms: i) Drain induced barrier lowering, ii) Gate leakage. 5

(c) Implement the following expression using CMOS logic: $Y = AB + C(A + B)$. 5

(d) Find minimum number delay [*sic*: should be "minimum delay"] for the logic circuit below. Also find the gate size to achieve this delay. 5

(e) Find the stage effort for the ideal number of stages. 5



The figure of Term Test 2, question 4 (redrawn): inverter 6, then NOR2 4x and NAND3 3x, then NOR2 6y, NAND3 4y and NAND2 3y, then inverter z and the load 48; A and B are faint marks on the paper

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